



**NANYANG
TECHNOLOGICAL
UNIVERSITY**
SINGAPORE

Consult 5: CA6121 AI in Project Management

Group Project Progress

Presented by Dr Ravish Maqsood



Prepared by: Ravish Maqsood

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Project Procurement

Project Procurement



Definition

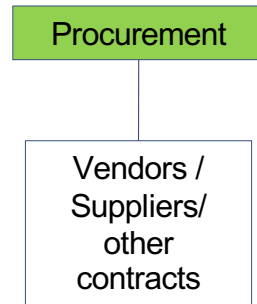
What is Procurement?

Project Procurement Management includes the processes necessary to **purchase or acquire products, services**, or results needed from outside the project team.

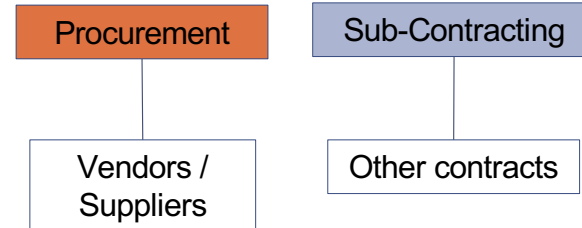


Procurement

Some companies use terminology
“**Procurement**” only for purchasing of
Equipment and other items

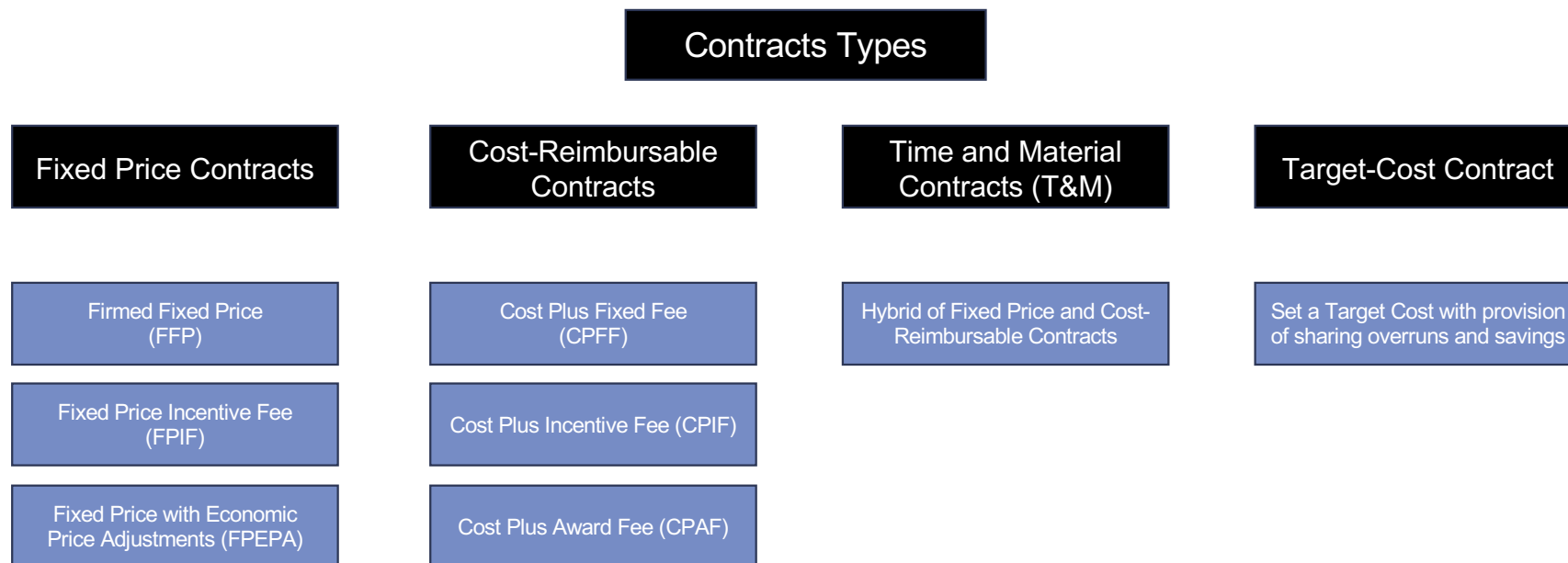


Some companies use terminology
“**Procurement**” to combine purchasing of
Equipment and other items PLUS sub-
contracts



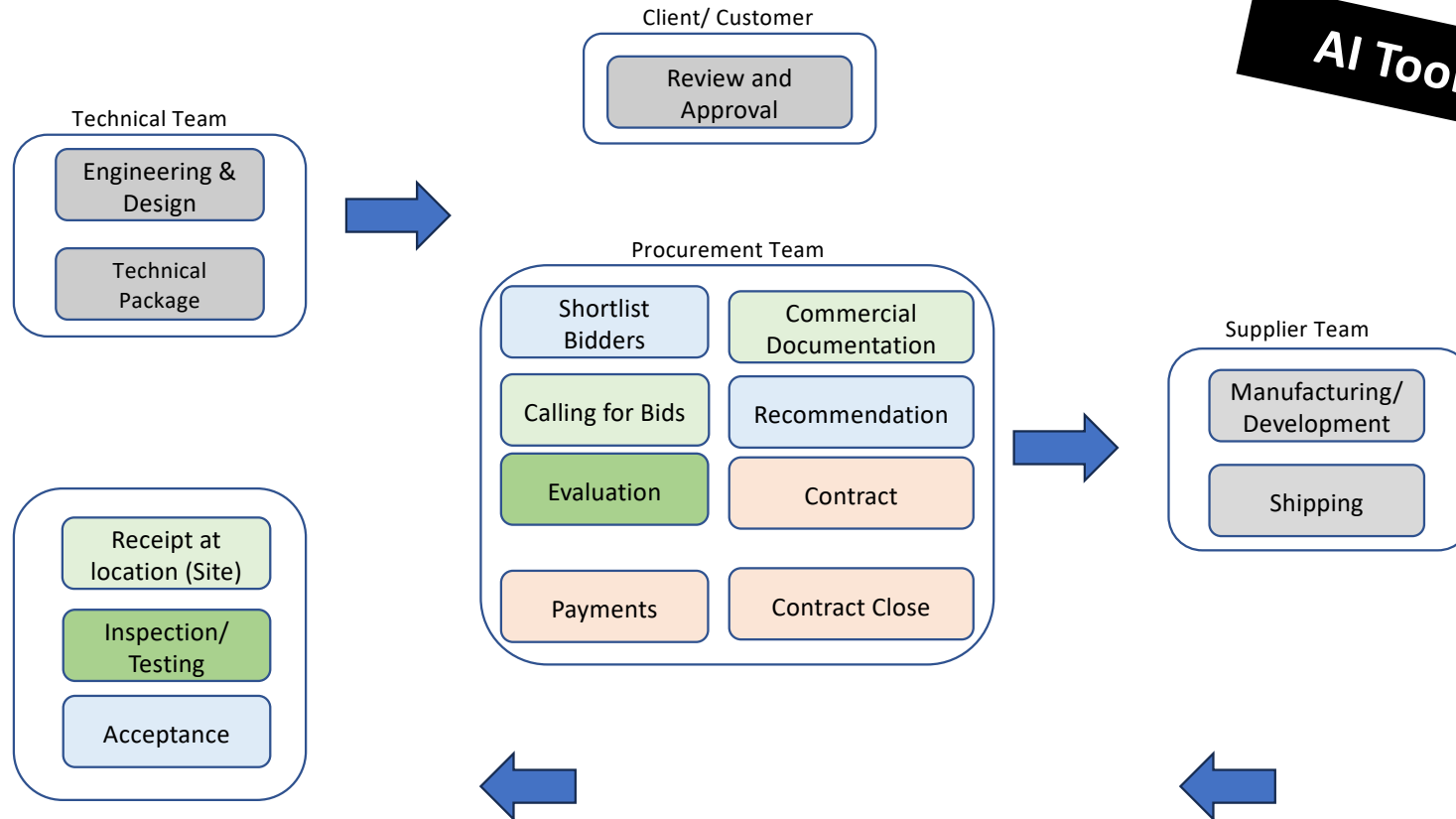


Plan Procurement Management





Procurement Process Cycle



AI Tool can help



Procurement: Pre-Order and Post Order Activities

AI Tool can help

Procurement Prime Responsibility:

Purchase Equipment, items, place orders and ensure its timely delivery to the project

Procurement Activities

Pre-Order/Award Activities

- Vendor Selection
- Preparing Bid / Tender Documents
- Calling Bids/ Tenders
- Evaluating
- Award

Post-Order/Award Activities

- Order tracking and coordination
- Expediting and Inspection
- FAT (Factory Acceptance Test) witness
- Shipping and Delivery to the site
- Material Control and Management
- Over, Short and Damaged (OS&D)
- Contract close out



Procurement Risks

Project Procurement Management



Procurement Risks: Vendors / Suppliers / Sub-Contractors

AI Tool can help

Timely Approvals:

- Can we proceed without client's approvals?
- Any client suggested vendor/ supplier/ sub-contractor/ specialist must be recorded and duly signed by the client.
- Such list must be mentioned in PMP (Project Management Plan).
- Past performance and financial checks still must be performed on client's suggested vendor/ supplier/ sub-contractor/ specialist.
- Single source items and vendor/ supplier/ sub-contractor/ specialist must be recorded and duly signed.



Procurement Risks

Risks need to be managed:

- Financial failure of the Vendor / Supplier/ Sub-contractor
- Under performance
- Delays (Development, Fabrication, Delivery, Installation etc.)

AI Tool can help



Project Quality

Project Quality Management



Definition

What is Quality???



Definition

Meeting customer's expectations and perceptions





Project Risk Assessment

Project Risk Management



Definition

What is Risk?

Project Risk is an **uncertain event or condition** that, if occurs,
has a positive or negative **effect on** one or more project
objectives such as
scope, schedule, cost and quality.



Risks

- Risks will continue to emerge through out project lifecycle.
- Risks must be monitored and managed.
- Important to manage the risks to deliver the project within the expectations of contractual agreement.
- Level of acceptable level of risk exposure to be agreed and known upfront.
- Contingencies are built in the cost and schedule estimates for risks.



Risk Classification

Unknown-Known (Hidden fact) Knowledge exists in the community but not with the entity working on the endeavor.	Unknown-Unknown (Emergent risk) Knowledge does not exist within the sphere of influence.
Known-Known (Facts and requirements) Managed as a part of scope. Not a risk.	Known-Unknown (Classic risk) There is knowledge to identify probability and impact.

Figure 2-46. Risk Classification



**How much % of group project work
have your group completed so far?**

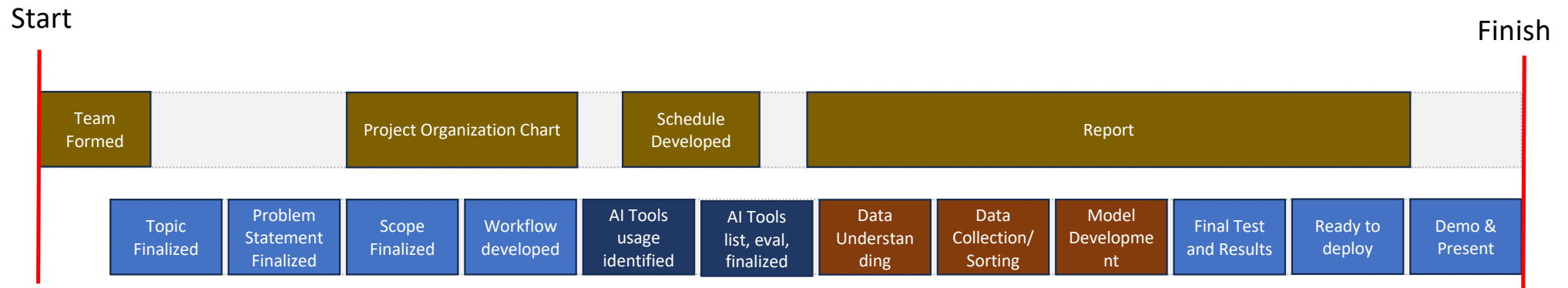


Your Group Project Progress?

AI in Project Management



Group Project Approach



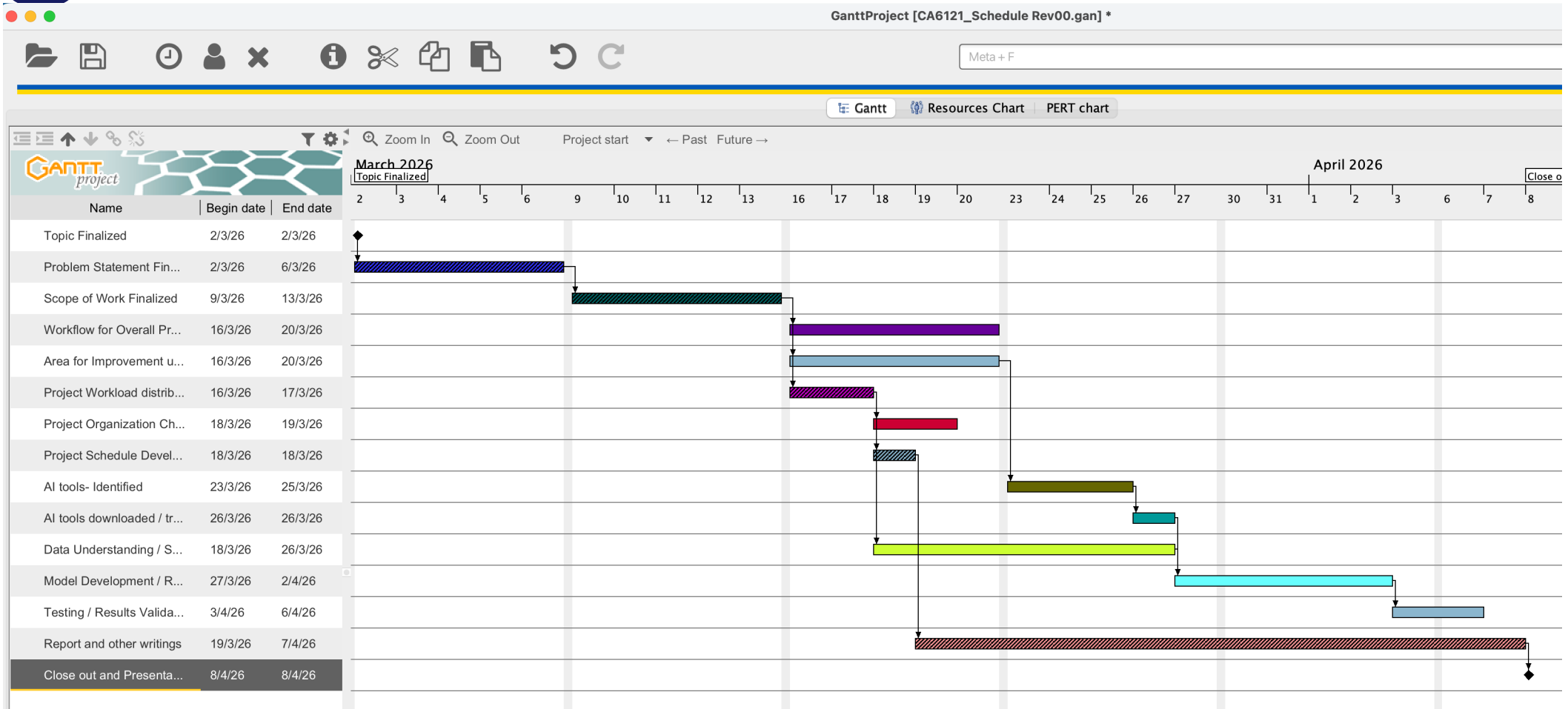


Project Steps & Planning- Gantt Chart

Project Execution Steps: For Reference only					
S.No	STEPS	Resources	Involvement	Duration	Predecessor
1	Topic Finalised	1,2,3,4,5 (All Team)	20% time of each person	1	Nil
2	Problem statement finalised	1,2,3,4,5 (All Team)	20% time of each person	5	Topic Finalised
3	Scope of work finalised	1,2,3,4,5 (All Team)	20% time of each person	5	Problem statement finalised
4	Workflow for the overall project developed	1,2,3,4,5 (All Team)	20% time of each person	5	Scope of work finalised
5	Areas for improvement using AI tools in workflow identified	1,2,3,4,5 (All Team)	20% time of each person	5	Scope of work finalised
6	Workload distributed among the team	1 (Project Manager)	20% time	2	Scope of work finalised
7	Project Organization chart is developed	1 (Project Manager)	20% time	2	Workload distributed among the team
8	Project Schedule Developed for the overall project	4 (Planning / Sch / Cost Expert)	20% time	1	Workload distributed among the team Project Organization chart is developed
9	AI Tools to be used identified, listed, evaluated, and finalized	2,3 (Data and AI Tools Expert)	20% time of each person	3	Areas for improvement using AI tools in workflow identified
10	AI Tools downloaded / installed / subscribed (free version).	3 (AI tools Expert)	10% time	1	AI Tools to be used identified, listed, evaluated, and finalized
11	Data understanding, Collection, Sorting completed for the project	1,2,3,5 (All team except Planning/Sch/cost)	10% time Project Manager 20% time Data Expert 20% time AI Tools expert 10% time Testing/ QA/ Deployment expert	7	Workload distributed among the team
12	Model development / Run / Testing Complete to use the data	1,2,3,5 (All team except Planning/Sch/cost)	20% time of each person	5	AI Tools downloaded / installed / subscribed Data understanding, Collection, Sorting completed for the project
13	Testing / Results are validated / Ready to Deploy	2,3,5 (Data & AI Tool Expert, Testing Expert)	100% time Data Expert 10% time AI Tools expert 20% time Testing/ QA/ Deployment expert	2	Model development / Run / Testing Complete to use the data
14	Report Writing : Benefits/ Challenges / Lessons for these AI tools on this project identified and recorded	1,2,3,4,5 (All Team)	20% time Project Manager 10% time Data Expert 10% time AI Tools expert 20% time Planning / Sch / Cost Expert 10% time Testing/ QA/ Deployment expert	14	Project Schedule Developed for the overall project



Project Steps & Planning- Gantt Chart



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Project Topics / Presentation Sequence / Important Dates



Project Topics

S.No	Group No.	Project Title (Short Description)
1	1, 3, 19, 20,21, 13	Job Application
2	2, 10, 16	Notes Of Meeting
3	11, 12	Contract Document Summary
4	4	AI-Powered Agile Requirements Engineering: Intelligent User Story and Acceptance Criteria Generation System
5	5	Issue tracker (Improvement)
6	6	What is in my fridge
7	7	AI-Assisted Event Synchronization for Construction Project Management
8	8	Personalized Periodic Health Check-up
9	9	Keeping track of product failures using AI to help in performing the analysis to improvement.
10	14	Project Feasibility Analysis for Startup
11	15	Weekly posts for the team communication (key messages, updates, announcement, etc.).
12	17	Managing AI Education Product Development: Leveraging AI Models and Agents for Enhanced Project Management
13	18	Analyzing data set to propose the best fit solution for any process of your choice.
14	22	Resources allocation on a game development project.
15	23	AI-Driven Personalized Scheme Generation and Intelligent Full-Process Management of Multi-Party Communication for Smart Home Appliance Planning and Deployment of Renovation Groups



Group Selection for Presentation

8 groups
to be selected

Spin the

!!! Wheel !!!

SATURDAY



Group Selected for Presentation

Sequence	Group No.	Project Title (Short Description)
	1, 3, 19, 20,21, 13	Job Application
	2, 10, 16	Notes Of Meeting
	11, 12	Contract Document Summary
	4	AI-Powered Agile Requirements Engineering: Intelligent User Story and Acceptance Criteria Generation System
	5	Issue tracker (Improvement)
	6	What is in my fridge
	7	AI-Assisted Event Synchronization for Construction Project Management
	8	Personalized Periodic Health Check-up
	9	Keeping track of product failures using AI to help in performing the analysis to improvement.
	14	Project Feasibility Analysis for Startup
	15	Weekly posts for the team communication (key messages, updates, announcement, etc.).
	17	Managing AI Education Product Development: Leveraging AI Models and Agents for Enhanced Project Management
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	22	Resources allocation on a game development project.
	23	AI-Driven Personalized Scheme Generation and Intelligent Full-Process Management of Multi-Party Communication for Smart Home Appliance Planning and Deployment of Renovation Groups



FINAL QUIZ & PRESENTATIONS - Schedule

DAY 1

11April2026 Saturday (1:30pm- 2:15pm)

	Start	End	FINAL QUIZ
1	1:30pm	2:15pm	35 min (45min for special case)

DAY 1

11April2026 Saturday (2:30pm- 5:00pm)

	Start	End	Group to present: At random picked up
1	2:30pm	2:45pm	
2	2:45pm	3:00pm	
3	3:00pm	3:15pm	
4	3:15pm	3:30pm	
5	3:30pm	3:45pm	
6	3:45pm	4:00pm	
7	4:00pm	4:15pm	
8	4:15pm	4:30pm	



Important dates and timeline: GROUP REPORT

- **Group report** (*.ppt format only) to be submitted by: **09 April 2026 (8pm)**
- File name as **GroupNN_report.ppt**
- Please **submit your report via NTU Learn** folder provided.
- **Only one student** from each group **to submit** one copy.
- Please also email a copy of ***.ppt format** submission to ravish.maqsood@ntu.edu.sg
- Mark subject of the email: **CA6121-GroupNN-Presentation Submission.**

Note: No submission after 2hours buffer time shall be entertained.



Important dates and timeline: FINAL QUIZ

- **Final Quiz: 11 April 2026 (1:30pm)**
- Online quiz on NTU Learn.
- Please set up your **lockdown browser** on your laptop and test prior coming to the class.
- MCQ (Multiple Choice Question) format.
- Closed Book
- Closed Notes
- Attendance will be marked for this compulsory quiz. (No make-up quiz is planned).



Important dates and timeline: PEER REVIEW

- **Peer Review completion between:**

03 April 2026 (1:00am) to 11 April 2026 (11:59pm)

- Online peer review via NTU Learn.
- Please ensure you complete it on time as penalty will be imposed for not completing on time.
- No extension of time is given.

Do not edit
How to change the design



Audience Q&A

① The Slido app must be installed on every computer you're presenting from

slido



Project Progress



Your Group Project Progress?

1. Teamwork is important.
2. Group lead acting as project manager is reminded to coordinate group assignment activities.
3. Group assignment progress is monitored and managed.
 - Problem Statement
 - Scope
 - Workflow development
 - Tools selected and work is in progress



Any questions?



Progress Check



Get Ready for your draft work

Recap

- Any group can be asked to about their group work progress.
- It could be discussion on their project report draft.
- At random any of the groups can be picked to share.



What to do before next consultation?



Get Ready

- Ensure to watch all the lessons.
- Attempt the sample quiz questions to familiarize.
- Setup lockdown browser on your laptops for the final quiz.
- Work towards completion of your project report and work presentation.
- Check your work prior getting ready to submit.



What to do before next tutorial class?



Tutorial Classes

- Ensure you have **watched the relevant lecture(s)**.
- Familiarize yourself with
 - **Risk management and quality related concepts on a project**
- Please **bring your laptop** to participate in class exercises/ activities.



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End of class

